BioHackrXiv

BioHackEU23 report: Template for the very long title

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Introduction

Results

Proposed Use Cases

Proposed Use Cases

Bashir: I propose three use cases using NORMAN-SLE, ExpoPortal ontologies, and scientific publications, initially focusing on PFAS.

Bashir: Use case 1. Ontology Gap Detection and Improvement

Question: Can scientific publications help identify missing concepts and relationships in exposome ontologies?

Approach: - Extract chemical, exposure, and health concepts from scientific publications using AI. - Compare extracted information with ontologies in ExpoPortal. - Identify missing concepts, relationships, and cross-ontology mappings. ###the first draft of the result is available on the : https://github.com/egonw/biohackeu26-project19/blob/main/paper/Bashir_usecase_1.md

Expected Output: Potential ontology gaps and evidence-based suggestions for improvement.

Bashir: Use case 2. Scientific Research Gap Explorer

Question: Which chemicals and exposure-related health effects need further scientific research?

Approach: - Explore NORMAN-SLE chemicals and related concepts using ExpoPortal ontologies. - Analyze scientific publications to assess available evidence. - Identify topics with limited, missing, or conflicting evidence.

Expected Output: Potential research gaps and suggested research questions. ### the first draft of the result is available on the : https://github.com/egonw/biohackeu26-project19/blob/main/paper/Bashir_usecase_2.md ### Bashir: Use case 3. Chemical Exposure & Health Knowledge Graph

Question: How can we connect chemicals, exposure pathways, and health effects using multiple ontologies?

Approach: - Link NORMAN-SLE chemicals to relevant ExpoPortal ontologies. - Explore environmental sources, exposure pathways, biological processes, and health outcomes. - Connect relationships to supporting publications.

Expected Output: An interactive knowledge graph with traceable scientific evidence.

Fernando: Use case 4: Impact of Air Pollution on Health Outcomes and Transcriptomic/Proteomic Profiles

- Integrate phenotypic, exposure, demographic, transcriptomic, and proteomic data.
- Map concepts/entities to ExpoPortal ontologies.
- Identify ontology gaps after mapping.
- Apply neuro-symbolic AI to identify associations between the expomosome and BMI/asthma or omic expression levels.

Discussion

Bashir: Evaluation and Future Work

- Evaluate ontology coverage and mappings.
- Discuss limitations and missing evidence.
- Explore dataset and publication integration.
- Propose future ExpoPortal improvements. ...

Acknowledgements

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References

Author contributions

First Author: Writing – original draft; Last Author: Conceptualization, Writing – review & editing.

Appendices

If you want the Appendix (-ces) to show up after the references, wrap them in after the header, like done in this Markdown file. Look at the source to see the exact structure.